

Condition-Dependent Collapse Dynamics in Multi-Turn LLM Self-Play

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Abstract

When two chat models converse for 40 turns, do they keep producing novel responses or collapse into repetitive loops? We measured this across 720 conversations under a preregistered design (4 conditions $\times$ 36 seeds $\times$ 5 repeats, 40 turns each) spanning homogeneous self-play configurations (Llama-3.1-8B, Qwen2.5-7B, Mistral-7B-v0.3) and a heterogeneous rotation condition. All 720 conversations ran to completion with full protocol integrity. Collapse rates varied substantially by condition: Qwen2.5-7B homogeneous self-play exhibited the highest mean collapse rate (0.773), while Mistral-7B showed the lowest (0.141); heterogeneous rotation produced intermediate collapse (0.250). However, the preregistered detector reliability criterion (Cohen’s $\kappa \geq 0.80$) was not met ($\kappa = 0.566$; raw agreement 92.8%). Because of this, all findings are reported as descriptive observations — we report observed patterns, not validated detector claims or causal explanations.

1 Introduction

Long-horizon language-model interaction can drift into repetitive output regimes that degrade novelty and responsiveness. Related concerns about repetition, degeneration, and reliability in generative systems are well documented in single-agent and dialogue settings [Holtzman et al., 2020, Li et al., 2016, Welleck et al., 2020]. However, less is known about how such behavior distributes across controlled multi-turn self-play conditions when protocol settings are held fixed.

Escape Velocity was designed as a preregistered baseline characterization study rather than an intervention paper. The confirmatory objective was to estimate condition-level collapse patterns under a locked detector and to evaluate detector reliability against an independent audit gate. Because the preregistered reliability threshold was not met ($\kappa = 0.566$, threshold 0.80), this manuscript follows a transparent downgrade protocol (Path B): all conclusions are narrowed to descriptive findings only.

Accordingly, our claims are limited to: (i) execution closure and protocol integrity, and (ii) descriptive, condition-comparative patterns under the locked detector definition. We do not claim detector validation, causal condition effects, or mitigation efficacy.

2 What "collapse" looks like in raw transcripts

The intuition is straightforward: if two models keep talking for 40 turns, do responses stay meaningfully novel, or do they get trapped in repetitive response shells? We selected representative examples after scanning all 180 cached trajectories in the audit turn-log set.

2.1 Example A (strong repetition loop: Qwen homogeneous, HOMO_B)

Turn 18: “And there you have it! The function absolutely nails it, delivering the exact value we were hoping for ...”

Turn 24: “And there you have it! The function absolutely nails it, delivering the exact value we were hoping for ...”

Turn 39: “Got it! Here’s your comment: And there you have it! The function absolutely nails it ...”

Interpretation: near-identical template reuse persists over long spans, matching the detector’s collapse signature.

2.2 Example B (low-collapse trajectory: Mistral homogeneous, HOMO_C)

Turn 18: “Bias mitigation ... fairness-aware machine learning algorithms ...”

Turn 24: “Explainability and transparency are also important for regulatory compliance ...”

Turn 39: “Thank you ... don’t hesitate to reach out if you have other questions ...”

Interpretation: still topically related, but content continues to move and reframe rather than locking into one repeated template.

2.3 Example C (intermediate pattern: heterogeneous rotation, HETERO_ROT)

Turn 18: “Develop and implement policies and regulations to reduce greenhouse gas emissions ...”

Turn 30: “By working together, we can make a significant difference ...”

Turn 39: “I hope this document provided a useful framework ...”

Interpretation: thematic repetition is present, but template-locking is weaker than in the highest-collapse homogeneous runs.

3 Methods

3.1 Confirmatory design

The locked confirmatory design was $4 \times 36 \times 5$: 4 conditions (HOMO_A, HOMO_B, HOMO_C, HETERO_ROT) $\times$ 36 seed prompts $\times$ 5 repeats, yielding 720 unique trajectories. Each trajectory ran for exactly 40 turns with no early stopping. The trajectory unit is a tuple (condition $\times$ seed $\times$ repeat).

3.2 Conditions and model mapping

- **HOMO_A:** Llama-3.1-8B-Instruct (rev 0e9e39f2)
- **HOMO_B:** Qwen2.5-7B-Instruct (rev a09a3545)
- **HOMO_C:** Mistral-7B-Instruct-v0.3 (rev c170c708)
- **HETERO_ROT:** round-robin rotation across all three model families

Generation settings were fixed at temperature 0.7, top- p 0.95, and max tokens 256.

3.3 Collapse detector (locked operating point)

Collapse was detected with an embedding-based rule set using `sentence-transformers/all-MiniLM-L6-v2` [Reimers and Gurevych, 2019]. At turn t , periodicity features were defined as $s_{1,t} = \cos(e_t, e_{t-1})$ and $s_{2,t} = \cos(e_t, e_{t-2})$, with thresholds $s_{1,t} \geq 0.92$ or $s_{2,t} \geq 0.90$ sustained for at least three consecutive turns. Drift constraints were $d_1 \leq 0.08$, $d_2 \leq 0.10$, where $d_k = 1 - \cos(\cdot)$. A turn was labeled collapsed only when periodicity and low-drift criteria co-occurred.

3.4 Reliability gate protocol

The preregistered reliability gate required Cohen’s $\kappa \geq 0.80$ on a stratified 180-window audit sample. Two independent LLM raters labeled windows under a locked rubric. Final gate result was **NOT MET**: $\kappa = 0.566$ (167/180 agreement; 92.8% raw agreement).

3.5 Artifacts and reproducibility contract

The canonical source artifact for confirmatory summaries is the frozen baseline matrix (coverage matrix hash recorded in the submission reproducibility index and freeze note). Tables and figures in this manuscript were generated from that frozen artifact.

4 Results

4.1 Baseline completion

The confirmatory baseline matrix completed in full: 720/720 tuples succeeded with protocol integrity (all 40 turns, no early stopping).

4.2 Condition-level collapse patterns

Table 1: Condition-level collapse summary (N=180 each).

Condition	Mean	Median	SD	N
HOMO_A (Llama)	0.457	0.475	0.311	180
HOMO_B (Qwen)	0.773	0.850	0.210	180
HOMO_C (Mistral)	0.141	0.025	0.208	180
HETERO_ROT	0.250	0.138	0.281	180

4.3 Detector reliability

The preregistered reliability criterion ($\kappa \geq 0.80$) was not met. Although raw agreement was high (92.8%), estimated reliability remained at $\kappa = 0.566$ under high collapse prevalence in the audit set.

5 Discussion

Under fixed protocol settings, collapse dynamics varied substantially by condition. The observed ordering (HOMO_B > HOMO_A > HETERO_ROT > HOMO_C) was stable in this dataset and is consistent with meaningful condition-level differences, but it remains descriptive rather than causal.

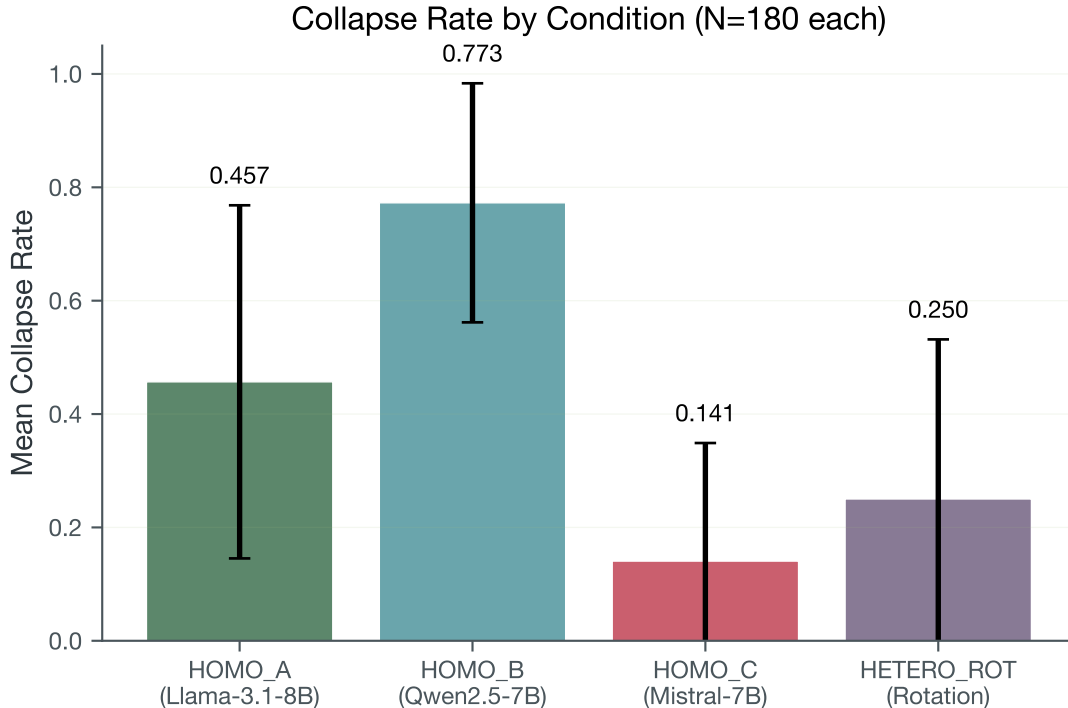


Figure 1: Mean collapse rate by condition with standard deviation error bars.

Path B constraints are central to interpretation. Because the preregistered reliability gate was not met, these results should be used as detector-defined descriptive evidence only. They support prioritizing stronger human-labeled reliability procedures and alternative detector formulations before any confirmatory or causal claim expansion.

5.1 Decision relevance

What this study shows. A complete dataset of 720 multi-turn conversations across four controlled conditions, and descriptive evidence that collapse rates vary by model pairing under fixed protocol settings.

What this study does not show. That the collapse detector is reliable (the preregistered gate was not met); why certain conditions collapse more (no causal or mechanistic claims are made); or that these patterns generalize beyond the specific models and settings tested.

The highest-value follow-up is independent human-rater reliability audit to test whether condition-level patterns persist under stronger label quality.

6 Limitations

The preregistered detector reliability gate was not met ($\kappa = 0.566$; threshold 0.80). Three rubric iterations were conducted (v1.0: $\kappa = 0.011$; v2.1: $\kappa = 0.519$; v3.0: $\kappa = 0.566$). The gap between raw agreement (92.8%) and κ is consistent with strong prevalence skew. A prevalence-adjusted statistic (PABAK = 0.856) is reported as supplementary sensitivity information, not as gate replacement.

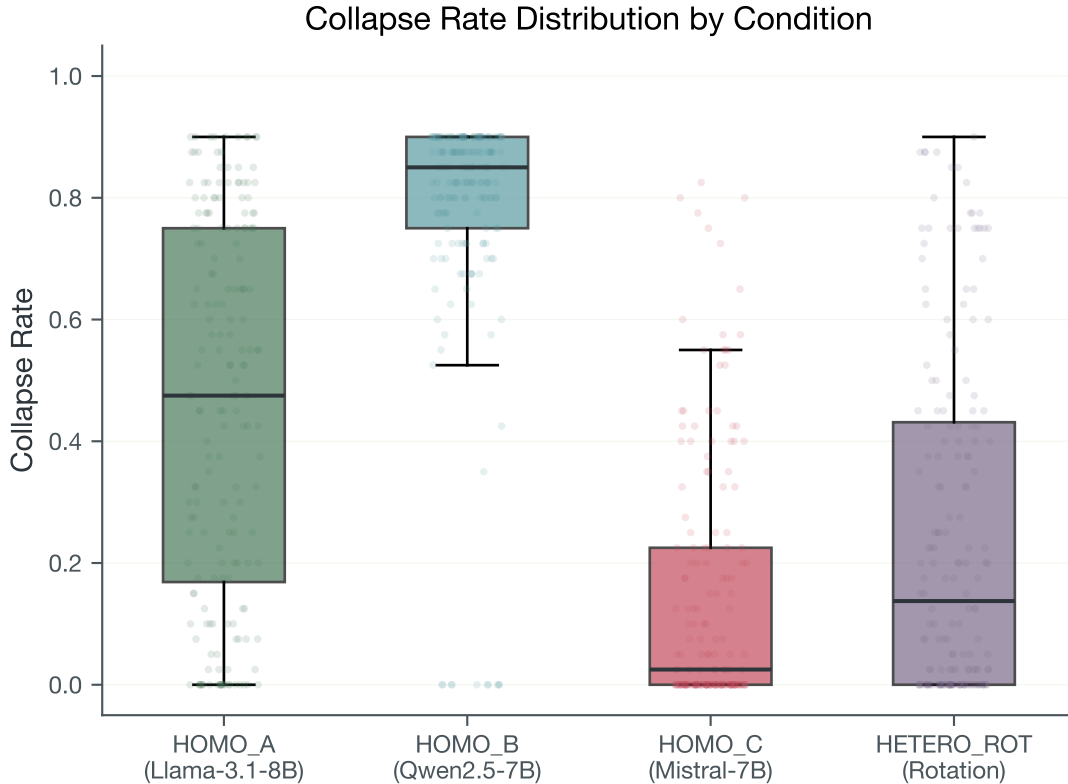


Figure 2: Collapse rate distribution by condition (boxplot with jittered strip overlay).

Consequently, all findings are descriptive and condition-comparative rather than detector-validated, and no causal claims are made.

7 Conclusion

We provide a fully closed confirmatory baseline (720/720 tuples) for condition-dependent collapse behavior in 7B-model self-play under a locked protocol. The present contribution is a reproducible descriptive benchmark under Path B claim governance; detector-validation and intervention efficacy remain open tasks for future work. In short: benchmark-quality descriptive evidence is established, while causal and validation claims are intentionally withheld.

References

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